THE DOMAIN OF DEFINITION OF A RIGHT DERIVED FUNCTOR AND ITS LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5139-5162, following the statement at 5111-5137.

Editorial note. Part (5) is the single compressed step: the multiplicative system axioms for S_E inherit from S via part (3) (arrow upgrade) and part (1) (vertices land in E); short mechanical chains, recorded here.

Editorial note. The final sentence 'A similar result holds for LF' follows the established mirror counting practice.

Allowed standard prerequisites (editorial).

- Closure and functoriality lemmas for RF. lemma-derived-inverts, lemma-2-out-of-3-defined (carded L26162), lemma-derived-shift, lemma-derived-functor (all same chapter, cited at ~5143-5148). Their statements are exactly parts (1)-(4): closure under shifts and triangles, functoriality, inversion of S.
- Universal property of the localization. lemma-universal-property-localization (same chapter, cited at ~5160). Factors RF through S_E^{-1}E as an exact functor (part (7)).
- Distinguished triangles in the localization. proof of proposition-construct-localization (same chapter, cited at ~5157). Characterizes distinguished triangles in S_E^{-1}E, giving exactness in part (6).

Source reference. Original reference 05Q8: focused-seeds/geometry/S9-der-derived-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SC: sources/stacks-derived/pinned-original-source.tex, source lines 4896–4896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SA: sources/stacks-derived/pinned-original-source.tex, source lines 4789–4789. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05SB: sources/stacks-derived/pinned-original-source.tex, source lines 4855–4855. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SU: sources/stacks-derived/pinned-original-source.tex, source lines 4874–4874. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SD: sources/stacks-derived/pinned-original-source.tex, source lines 5019–5019. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05R7: sources/stacks-derived/pinned-original-source.tex, source lines 1688–1688. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 05R6: sources/stacks-derived/pinned-original-source.tex, source lines 1577–1577. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Assumptions and notation as in Situation 05S8.*

(1) *The full subcategory $\mathcal{E}$ of $\mathcal{D}$ consisting of objects at which RF is defined is a strictly full triangulated subcategory of $\mathcal{D}$.*
(2) *We obtain an exact functor $RF : \mathcal{E} \longrightarrow \mathcal{D}'$ of triangulated categories.*
(3) *Elements of S with either source or target in $\mathcal{E}$ are morphisms of $\mathcal{E}$.*
(4) *Any element of $S_\mathcal{E} = Arrows(\mathcal{E}) \cap S$ is mapped to an isomorphism by RF.*
(5) *The set $S_\mathcal{E}$ is a saturated multiplicative system in $\mathcal{E}$ compatible with the triangulated structure.*
(6) *The functor $S_\mathcal{E}^{-1}\mathcal{E} \to S^{-1}\mathcal{D}$ is a fully faithful exact functor of triangulated categories.*
(7) *We obtain an exact functor*
$$RF : S_\mathcal{E}^{-1}\mathcal{E} \longrightarrow \mathcal{D}'.$$
(8) *If $\mathcal{D}'$ is Karoubian, then $\mathcal{E}$ is a saturated triangulated subcategory of $\mathcal{D}$.*

A similar result holds for LF.

Proof. Since S is saturated it contains all isomorphisms (see remark following Categories, Definition 05Q8). Hence (1) follows from Lemmas 05SB, 05SC, and 05SU. We get (2) from Lemmas 05SA, 05SU, and 05SC. We get (3) from Lemma 05SB. Part (4) follows from Lemma 05SB. Part (5) follows from the definitions and part (3). The fully faithfulness in (6) follows from (3) and the definitions. The fact that $S_{\mathcal{E}}^{-1}\mathcal{E} \to S^{-1}\mathcal{D}$ is exact follows from the fact that a triangle in $S_{\mathcal{E}}^{-1}\mathcal{E}$ is distinguished if and only if it is isomorphic to the image of a distinguished triangle in $\mathcal{E}$, see proof of Proposition 05R6. The factorization of $RF : \mathcal{E} \to \mathcal{D}'$ through an exact functor $S_{\mathcal{E}}^{-1}\mathcal{E} \to \mathcal{D}'$ follows from Lemma 05R7. Finally, part (8) follows from Lemma 05SD. □